

Day 13: Cloud Database Migration & Scalable Backend Hardening

Key Milestone: Neon Serverless PostgreSQL & Async Backend Deployment

- Sprint Focus:

Execution of core deliverables for Day 13 spanning AI/NLP Intent Classification, Email Header Forensics, IOC Intelligence, Relationship Graphing, Case Workflow, and Full-Stack Frontend.

Key Deliverables Completed:

Neon Cloud PostgreSQL Instance, Async SQLAlchemy Connection Layer, Database Performance Benchmarks.

Individual Team Member Task Logs & Technical Contributions

- Muniswami.Y — AI / NLP Lead

Domain Ownership:

Explainable AI, Intent & Phishing Detection, Social Engineering signals

Key Modules:

NLP/AI Engine, Explainability, Content Risk scoring

Day 13 Execution:

Hardened NLP model execution against adversarial payload injection and extremely long email header string overflows.
- Jeevan reddy.G — Header & Forensics Lead

Domain Ownership:

Email header analysis, SPF/DKIM/DMARC, Mail path reconstruction, Timestamp analysis

Key Modules:

Header Forensics, Mail Path, Infrastructure Risk

Day 13 Execution:

Configured SQLAlchemy async engine with `asyncpg` connection pooling and SSL mode sanitization for cloud PostgreSQL.
- Lahari.K — IOC & Infrastructure Lead

Domain Ownership:

URL, Domain, IP extraction, Geolocation, ASN/ISP mapping, Malicious indicator detection

Key Modules:

IOC Extraction, Geolocation, URL/Domain/IP Risk

Day 13 Execution:

Added database indexes on `origin\_ip`, `case\_id`, `sender\_domain`, and `created\_at` for high-speed cross-case searching.
- Hemanth.P — Graph & Campaign Intelligence Lead

Domain Ownership:

Attack Relationship Graph, Similarity detection, Campaign grouping & correlation

Key Modules:

Graph Engine, Campaign Detection, Similar Attack Detection

Day 13 Execution:

Optimized graph serialization algorithms to stream large nodal structures in sub-50ms JSON responses.
- Abbas.N — Risk, Case & Response Lead

Domain Ownership:

Dynamic Threat Score engine, Automatic Forensic Case creation, Recommended Response actions

Key Modules:

Risk Scoring, Case Management, Response Recommendations

Day 13 Execution:

Provisioned serverless Neon.tech PostgreSQL database (`threattrace\_db`) and executed production schema migrations.
- Moulaanbee.N — Frontend + Assistant + Simulator Lead

Domain Ownership:

Dashboard, Visualizations (Graph, Map, Timeline), AI Chatbot, Email Threat Simulator, Overall UI/UX

Key Modules:

Dashboard, AI Cybersecurity Assistant, Threat Simulator, Frontend Integration

Day 13 Execution:

Updated API client services to dynamically fallback between live cloud endpoints and local development mocks.

System Integration & Quality Verification

Cross-Module Integration: Successfully migrated all local SQLite records to live Neon Cloud PostgreSQL; verified zero connection pooling dropouts.

Quality Assurance & Verification: All technical modules, automated unit test suites, and API contracts for Day 13 executed with 100% clean passes. Code reviewed and merged into main repository.

Prepared By: Threat Trace AI Development Team | **Project Lead:** Muniswami.Y | **Status:** SPRINT DAY 13 MILESTONE SIGNED OFF